- For every Dapp project, a `<project>-app` module for the web application and a `<project>-contract` module for smart contracts
- A web server and a package manager (Node.js and the Node Package Manager [npm])
- A blockchain provider (such as Ganache) called web3 provider
- A development tool, the Truffle suite (IDE) that provides an integrated environment to deploy and test a Dapp
- Account management using the MetaMask browser plugin

The end-to-end development process introduced in this chapter will be expanded in future chapters with various other Dapps.

4.1 Dapp development using Truffle

Truffle is an integrated development environment and testing framework that provides a suite of features and commands for end-to-end Ethereum-based Dapp development, including commands for

- Initializing a template or base directory structure for a Dapp (`truffle init`)
- Compiling and deploying smart contracts (`truffle compile`)